[162060] Low-Resource Machine Translation

SoSe 2026

Language:	English	**Teacher:**	Christian Schuler
Place:	C7.2 - 1.05	**Office:**	D3 1 (Altbau) +1, Room 1.09
Time:	16:15-17:45	**E-mail:**	christian.schuler@dfki.de
Start:	14.04.2026	**Oberbabo:**	Simon Ostermann
Credits:	4 CP (R), 7 CP (R+H)		
Workload:	2 SWS		

Course Description

This seminar explores the challenges and possibilities of machine translation in contexts where data is scarce, incomplete, or entirely absent. Rather than focusing on standard, high-resource scenarios, the course examines what happens when core assumptions of modern NLP break down: when languages lack large corpora, established tools, or even a written tradition. Students will engage with both the technical and conceptual foundations of low-resource machine translation. Topics include data acquisition and corpus creation, evaluation under uncertainty, multilingual and transfer learning approaches, and strategies such as data augmentation, zero- and few-shot learning, and active learning. Alongside these methods, the course highlights the role of linguistic knowledge as a structured alternative to purely data-driven approaches. A central thread throughout the seminar is the critical examination of the broader context in which these systems are developed and deployed. Questions of ethics, bias, and responsibility are treated as integral to the field: whose languages are supported, how linguistic communities are represented, and what risks arise when systems are applied in sensitive or high-stakes domains. The course is intended for students with an interest in natural language processing, machine learning, and linguistic diversity, who want to understand both the limitations of current approaches and the strategies being developed to overcome them. It is particularly suited for those interested in research questions at the intersection of technology, language, and society.

Seminar, Seminar- What even is a Seminar?

"A seminar is a form of (...) recurring meetings, focusing each time on some particular subject, in which everyone present is requested to participate. This is often accomplished through an ongoing Socratic dialogue (...) It is essentially a place where assigned readings are discussed, questions can be raised and debates can be conducted." *

Seminar Schedule

Each session has a pool of papers, and we plan to have two presentations per session. Students present a paper of their choice (under the constraint that we need to fill each session) from the session's paper pool and lead the class discussion.

Some words on grading: This seminar is meant to be as interactive as possible. Final grades will be based on students' presentations, term papers (optional), but also on participation and discussion in class.

The participants are expected to prepare for classes accordingly, by reading the relevant papers and also doing background reading, if necessary. Based on this preparation, the participants should be able to discuss the presented papers in depth and to understand relevant context during the discussion.

*https://en.wikipedia.org/wiki/Seminar

Note: This preliminary syllabus may change in parts (e.g. list of papers to choose from).

Introduction, Course Overview, and Administrative Matters

"What are we going to do here?"

Session	Date	Content	Paper	Student
Phase 1 - Warm-Up				
1	14.04.2026	Administrative Session & Paper Pool		
2	21.04.2026	Introductory Lecture		
3	28.04.2026	Lingering Questions (optional)		
4	05.05.2026	Lingering Questions (optional)		
5	12.05.2026	LREC (no meeting)		
Phase 2 - Student's Paper Presentations				
6	19.05.2026	How to present and discuss	(Schuler et al., 2026)	C.S.
7	26.05.2026	Problem Space & Linguistic Foundations	(Kornai, 2013)	Q.M.
8	02.06.2026	Data Engineering & Augmentation	(Riabi et al., 2023)	A.K.
9	09.06.2026	Data Engineering & Augmentation	(Riabi et al., 2025)	J.W.
10	16.06.2026	Ethics & Policy	(Liu et al., 2022)	A.W.
11	23.06.2026	Evaluation & Quality Estimation	(Papineni et al., 2002)	N.G.
12	30.06.2026	Learning Paradigms & Architectural Methods	(Kholodna et al., 2024)	F.W.
13	07.07.2026	Emerging Directions & Open Problems	(Shompa, 2025)	P.H.
Phase 3 - Wrap-Up				
14	14.07.2026	Class Review & Synthesis		

Problem Space & Linguistic Foundations

"What defines low-resource languages and why do they pose unique challenges for MT?"

Digital Language Death
Kornai (2013) | **URL**

Survey of Low-Resource Machine Translation
Haddow et al. (2022) | **URL**

Speech Translation and the End-to-End Promise: Taking Stock of Where We Are
Sperber and Paulik (2020) | **URL**

The State and Fate of Linguistic Diversity and Inclusion in the NLP World
Joshi et al. (2020) | **URL**

ELLORA: Enabling Low Resource Languages with Technology
Bali et al. (2019) | **URL**

Cyber-geography of languages: Part 1: method, results and focus on English
Pimienta and de Oliveira (2022) | **URL**

Additional Reading

Language Death
Crystal (2000) | **URL**

Atlas of the world's languages in danger
Moseley and Nicolas (2010) | **URL**

Ethics & Policy

"Who benefits, who is harmed, and what governance is required?"

On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?
Bender et al. (2021) | **URL**

Challenges of Language Technologies for the Indigenous Languages of the Americas
Mager et al. (2018) | **URL**

Local Languages, Third Spaces, and other High-Resource Scenarios
Bird (2022) | **URL**

The State and Fate of Linguistic Diversity and Inclusion in the NLP World
Joshi et al. (2020) | **URL**

Decolonising Speech and Language Technology
Bird (2020) | **URL**

Not always about you: Prioritizing community needs when developing endangered language technology
Liu et al. (2022) | **URL**

Additional Reading

Data Engineering & Augmentation

"How to acquire, curate, or synthesize data under extreme scarcity?"

Europarl: A Parallel Corpus for Statistical Machine Translation
Koehn (2005) | **URL**

CCMatrix: Mining Billions of High-Quality Parallel Sentences on the Web
Schwenk et al. (2021) | **URL**

Findings of the WMT 2018 Shared Task on Parallel Corpus Filtering
Koehn et al. (2018) | **URL**

Seven Dimensions of Portability for Language Documentation and Description
Bird and Simons (2002) | **URL**

Massively Multilingual Sentence Embeddings for Zero-Shot Cross-Lingual Transfer and Beyond
Artetxe and Schwenk (2019) | **URL**

Improving Parallel Sentence Mining for Low-Resource and Endangered Languages
Okabe et al. (2025) | **URL**

Parallel Data, Tools and Interfaces in OPUS
Tiedemann (2012) | **URL**

The OPUS-MT Dashboard: A Toolkit for a Systematic Evaluation of Open Machine Translation Models
Tiedemann and de Gibert (2023) | **URL**

Improving Neural Machine Translation Models with Monolingual Data
Sennrich et al. (2016a) | **URL**

Understanding Back-Translation at Scale
Edunov et al. (2018) | **URL**

Exploring Diversity in Back Translation for Low-Resource NMT
Burchell et al. (2022) | **URL**

Beyond Dataset Creation: Critical View of Annotation Variation and Bias Probing of a Dataset for Online Radical Content Detection
Riabi et al. (2025) | **URL**

Treebanking user-generated content: a UD based overview of guidelines, corpora and unified recommendations
Sanguinetti et al. (2023) | **URL**

Enriching the NArabizi Treebank: A Multifaceted Approach to Supporting an Under-Resourced Language
Riabi et al. (2023) | **URL**

Additional Reading

Sub-word Modeling & Representation

"How to represent language when standard tokenization fails?"

SentencePiece: A Simple and Language Independent Subword Tokenizer and Detokenizer for Neural Text Processing
Kudo and Richardson (2018) | **URL**

Unsupervised Multilingual Learning for POS Tagging
Snyder et al. (2008) | **URL**

Unsupervised Learning of the Morphology of a Natural Language
Goldsmith (2001) | **URL**

Neural Machine Translation of Rare Words with Subword Units
Sennrich et al. (2016b) | **URL**

Modeling Language Variation and Universals: A Survey on Typological Linguistics for Natural Language Processing
Ponti et al. (2019) | **URL**

Saliency-driven Word Alignment Interpretation for Neural Machine Translation
Ding et al. (2019) | **URL**

Selective Sharing for Multilingual Dependency Parsing
Naseem et al. (2012) | **URL**

Neural Machine Translation of Rare Words with Subword Units
Sennrich et al. (2016b) | **URL**

LINSPECTOR: Multilingual Probing Tasks for Word Representations
Şahin et al. (2020) | **URL**

Zero-Shot Neural Machine Translation with Self-Learning Cycle
Lakew et al. (2021) | **URL**

A Rule-Based Kurdish Text Transliteration System
Ahmadi (2019) | **URL**

Measuring Similarity by Linguistic Features rather than Frequency
Delmonte and Busetto (2022) | **URL**

Additional Reading

Evaluation & Quality Estimation

"How to measure MT quality reliably when references and benchmarks are scarce?"

BLEU: A Method for Automatic Evaluation of Machine Translation
Papineni et al. (2002) | **URL**

A Structured Review of the Validity of BLEU
Reiter (2018) | **URL**

BLEURT: Learning Robust Metrics for Text Generation
Sellam et al. (2020) | **URL**

COMET: A Neural Framework for MT Evaluation
Rei et al. (2020) | **URL**

Additional Reading

Domain Adaptation & Moadlities

"What happens to machine translation when the language was never primarily a text?"

Analyzing phonetic variation in the traditional English dialects: Simultaneously clustering dialects and phonetic features
Wieling et al. (2013) | **URL**

Additional Reading

Learning Paradigms & Architectural Methods

"How to train or adapt models effectively with minimal supervision?"

Google's Multilingual Neural Machine Translation System: Enabling Zero-Shot Translation
Johnson et al. (2017) | **URL**

Massively Multilingual Neural Machine Translation
Aharoni et al. (2019) | **URL**

Unsupervised Cross-lingual Representation Learning at Scale
Conneau et al. (2020) | **URL**

Transfer Learning across Low-Resource, Related Languages for Neural Machine Translation
Nguyen and Chiang (2017) | **URL**

Polylingual Topic Models
Mimno et al. (2009) | **URL**

Ensemble Learning for Multi-source Neural Machine Translation
Garmash and Monz (2016) | **URL**

Attention is All You Need
Vaswani et al. (2017) | **URL**

Neural Machine Translation
Koehn (2020) | **URL**

Unsupervised Neural Machine Translation
Artetxe et al. (2018) | **URL**

Neural Machine Translation through Active Learning on Low-Resource Languages: The Case of Spanish to Mapudungun
Pendas et al. (2023) | **URL**

LLMs in the Loop: Leveraging Large Language Model Annotations for Active Learning in Low-Resource Languages
Kholodna et al. (2024) | **URL**

Zero-Shot Neural Machine Translation with Self-Learning Cycle
Lakew et al. (2021) | **URL**

Ensemble Learning for Multi-source Neural Machine Translation
Garmash and Monz (2016) | **URL**

A Survey of Domain Adaptation for Neural Machine Translation
Chu and Wang (2018) | **URL**

Finding the Right Recipe for Low Resource Domain Adaptation in Neural Machine Translation
Adams et al. (2022) | **URL**

On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?
Bender et al. (2021) | **URL**

Additional Reading

No Language Left Behind: Scaling Human-Centered Machine Translation
Team et al. (2022) | **URL**

Emerging Directions & Open Problems

"What new paradigms and unresolved issues will shape the near future?"

Should All Cross-Lingual Embeddings Speak English?
Anastasopoulos and Neubig (2020) | **URL**

Natural Language Processing for Dialects of a Language: A Survey
Joshi et al. (2025) | **URL**

Large Multimodal Models for Low-Resource Languages: A Survey
Lupascu et al. (2025) | **URL**

Neural Sign Language Translation
Camgoz et al. (2018) | **URL**

The Tatoeba Translation Challenge - Realistic Data Sets for Low Resource and Multilingual MT
Tiedemann (2020) | **URL**

Low-Resource Speech Translation
Bansal (2019) | **URL**

Sign Language Translation: A Survey of Approaches and Techniques
Liang et al. (2023) | **URL**

Speech Translation and the End-to-End Promise: Taking Stock of Where We Are
Sperber and Paulik (2020) | **URL**

A Practical Tool to Help Automate Interlinear Glossing: a Study on Mukr Kurdish
Asadpour et al. (2025) | **URL**

Learning to Recognize Dialect Features
Demszky et al. (2021) | **URL**

Multi-VALUE: A Framework for Cross-Dialectal English NLP
Ziems et al. (2023) | **URL**

Translation Asymmetry in LLMs as a Data Augmentation Factor: A Case Study for 6 Romansh Language Varieties
Vamvas et al. (2026) | **URL**

Versteasch du mi? Computational and Socio-Linguistic Perspectives on GenAI, LLMs, and Non-Standard Language
Platzgummer et al. (2026) | **URL**

Can Character-based Language Models Improve Downstream Task Performances In Low-Resource And Noisy Language Scenarios?
Riabi et al. (2021) | **URL**

Can Multilingual Language Models Transfer to an Unseen Dialect? A Case Study on North African Arabizi
Muller et al. (2020) | **URL**

Large Multimodal Models for Low-Resource Languages: A Survey
Lupascu et al. (2025) | **URL**

Disentangling meaning from language in LLM-based machine translation
Lasnier et al. (2026) | **URL**

Additional Reading

Proceedings of the First Workshop on NLP applications to field linguistics
Serikov et al. (2022) | **URL**

Proceedings of the Second Workshop on NLP Applications to Field Linguistics
Serikov et al. (2023) | **URL**

Proceedings of the Third Workshop on NLP Applications to Field Linguistics
Serikov et al. (2024) | **URL**

Proceedings of the Fourth Workshop on NLP Applications to Field Linguistics
Le Ferrand et al. (2025) | **URL**

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